Windows EVTX Log Analysis

Using EvtxECmd and Timeline Explorer

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Abstract

This report serves as a practical tutorial for analysts who are new to Windows event log forensics. It demonstrates how to configure Windows audit policies through the Local Security Policy editor (secpol.msc) so that a workstation generates the appropriate security events, how to parse those events from a .evtx binary log file into a structured CSV using Eric Zimmerman's EvtxECmd tool, and how to load and filter that output in Timeline Explorer for rapid investigation. Three categories of security-relevant events are examined in detail: successful and failed user logon and logoff activity (Event IDs 4624, 4625, and 4634), new local account creation (Event ID 4720), and the addition of a user to the local Administrators group (Event ID 4732). Understanding these events and how to surface them quickly is a foundational skill in host-based forensics, allowing an analyst to reconstruct who accessed a system, when, and what privilege changes occurred.

Background

How .evtx Files Are Generated

Windows records security-relevant activity through the Windows Event Log service, which writes structured binary XML records to files stored in C:\Windows\System32\winevt\Logs\. The Security event log, Security.evtx, captures events related to authentication, account management, and privilege use. Each record contains a fixed system header (EventID, TimeCreated, Computer, Channel) and a variable EventData payload whose fields differ depending on the event type. Because the binary .evtx format is not human-readable, a parser is required before the data can be analyzed efficiently.

Running EvtxECmd

EvtxECmd, developed by Eric Zimmerman and part of his open-source DFIR tools collection, parses .evtx files and normalizes the output into a flat CSV. It uses a Maps system to translate each event ID's unique EventData payload into standardized columns such as UserName, RemoteHost, and PayloadData1-6. This approach means all event types are represented in a single CSV rather than requiring separate files per event type.

To parse the provided security2.evtx file, PowerShell was opened as Administrator, the working directory was changed to the EvtxECmd folder inside ZimmermanTools, and the following command was executed:

```
.\\EvtxECmd.exe -f "C:\\Users\\student\\Desktop\\security2.evtx" --csv "C:\\
Users\\student\\Desktop" --csvf security2_parsed.csv
```

The -f flag specifies the input .evtx file, --csv specifies the output directory, and --csvf specifies the output filename. Once the command completes, EvtxECmd displays a summary including the total number of records parsed and a metrics table listing every event ID found in the file along with its count.

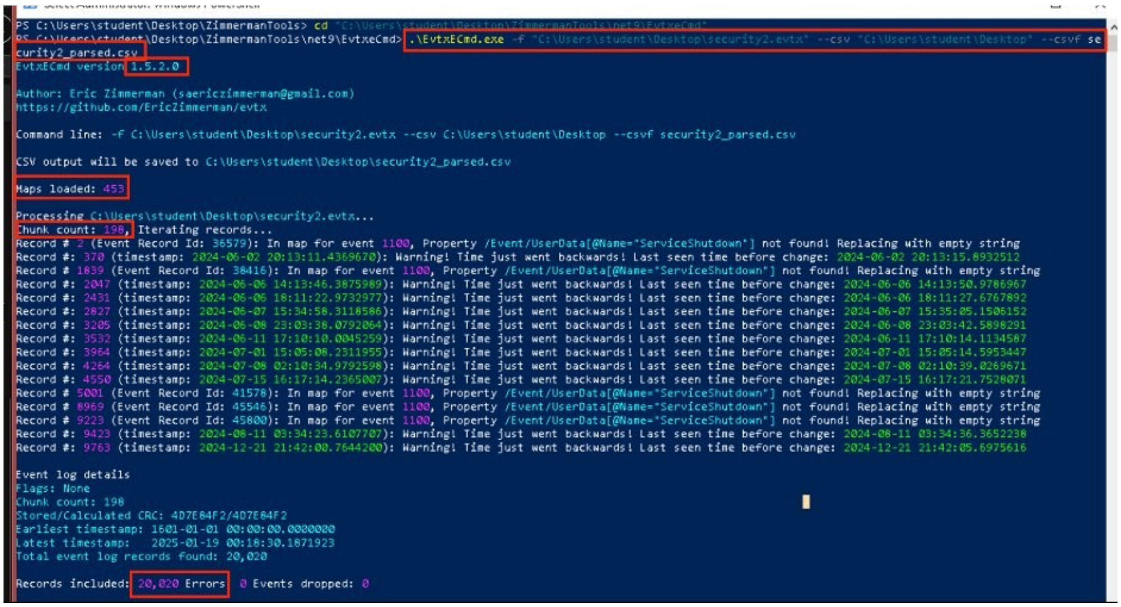

Figure 1. EvtxECmd v1.5.2.0 parsing security2.evtx, producing 20,020 records with 0 errors.

The metrics table confirms that all event IDs required for this assignment are present in the file: 4624 (1,230 occurrences), 4625 (2 occurrences), 4634 (24 occurrences), 4720 (1 occurrence), and 4732 (2 occurrences). The key event IDs are highlighted below.

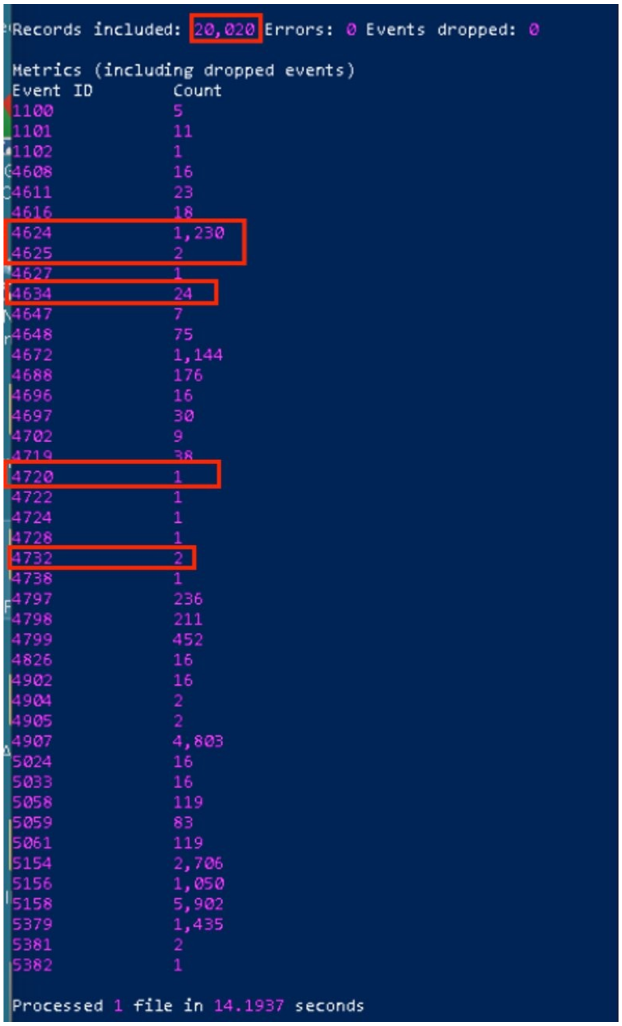

Figure 2. Event ID metrics output from EvtxECmd. Red boxes highlight the five event IDs investigated in this report.

Loading Results in Timeline Explorer

Timeline Explorer, also from Eric Zimmerman's suite, provides a graphical interface for loading and filtering the CSV output produced by EvtxECmd. After launching Timeline Explorer, the generated CSV is opened via File > Open. The tool renders each event as a row with columns including Time Created, Event Id, Map Description, User Name, Remote Host, PayloadData1 through PayloadData6, Provider, and Channel. The column header row includes filter boxes that allow an analyst to instantly isolate any event ID or user account of interest. The search bar at the top right provides a fast full-text search across all fields.

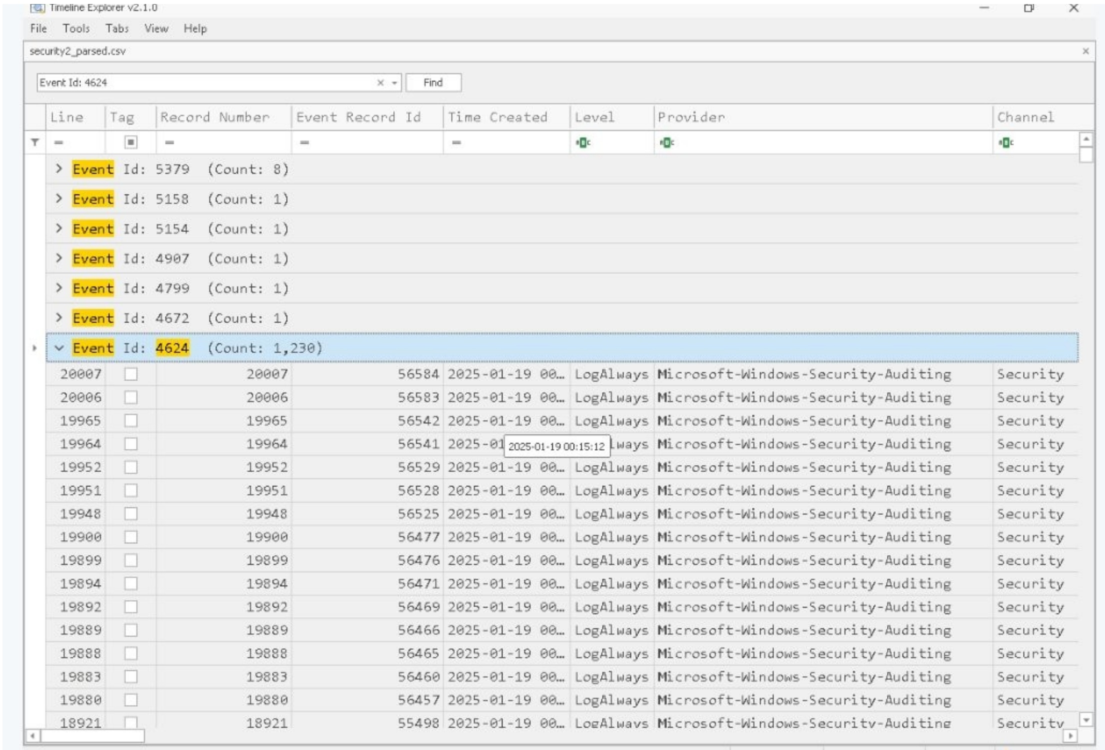

Figure 3. Timeline Explorer loaded with security2_parsed.csv, filtered to Event ID 4624 (Count: 1,230). The grouped view shows all event IDs that share search terms alongside the target events.

Section 1: Successful Login, Failed Login, and Logoff (Event IDs 4624, 4625, 4634)

Audit Policy Configuration

For Windows to generate logon and logoff events, the Audit logon events policy must be enabled. This is configured through the Local Security Policy editor, opened by pressing Windows + R and typing secpol.msc. Navigate to: Security Settings > Local Policies > Audit Policy > Audit logon events.

Double-clicking the policy opens its Properties dialog. Both the Success and Failure checkboxes should be enabled. Enabling Success causes the system to log Event ID 4624 on every successful interactive, network, or service logon, and Event ID 4634 on every logoff. Enabling Failure causes it to log Event ID 4625 whenever a logon attempt is rejected due to a bad password, unknown account, or policy restriction. Without this policy enabled, none of these events will appear in the Security log regardless of what activity occurs on the machine.

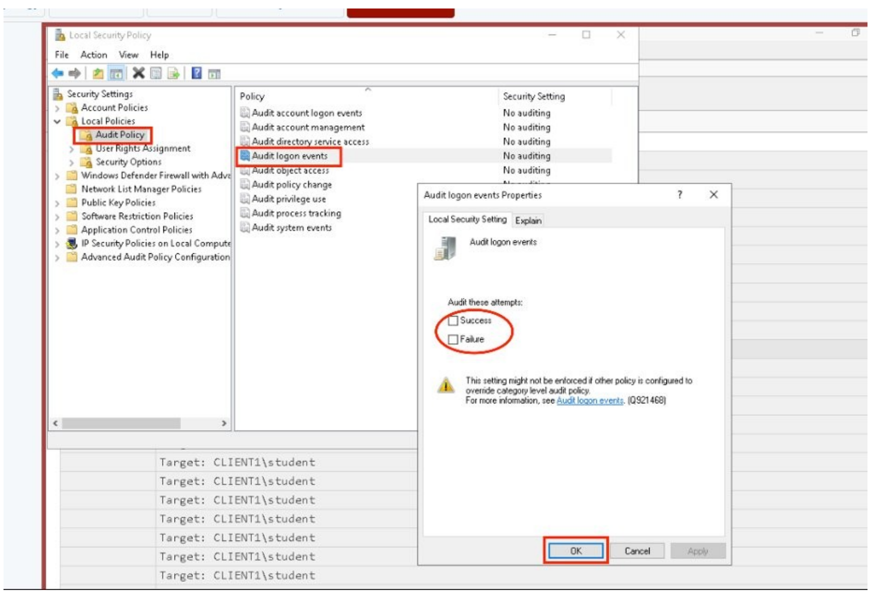

Figure 4. secpol.msc showing the Audit logon events policy highlighted under Local Policies > Audit Policy, with the Properties dialog open. Both Success and Failure must be checked to capture 4624, 4625, and 4634.

Event ID 4624: Successful Logon

Event ID 4624 is generated each time an account successfully logs on to the system. It is one of the most frequently occurring events in the Security log. In Timeline Explorer, filtering the Event Id column for 4624 reveals 1,230 records in the security2.evtx file. Scrolling right to the UserName, RemoteHost, and PayloadData1 columns provides the forensically relevant detail. The UserName column shows the account that initiated the logon (in domain\username format), the RemoteHost column shows the source workstation or IP address for network logons, and PayloadData1 contains the LogonType field, which identifies the method of authentication.

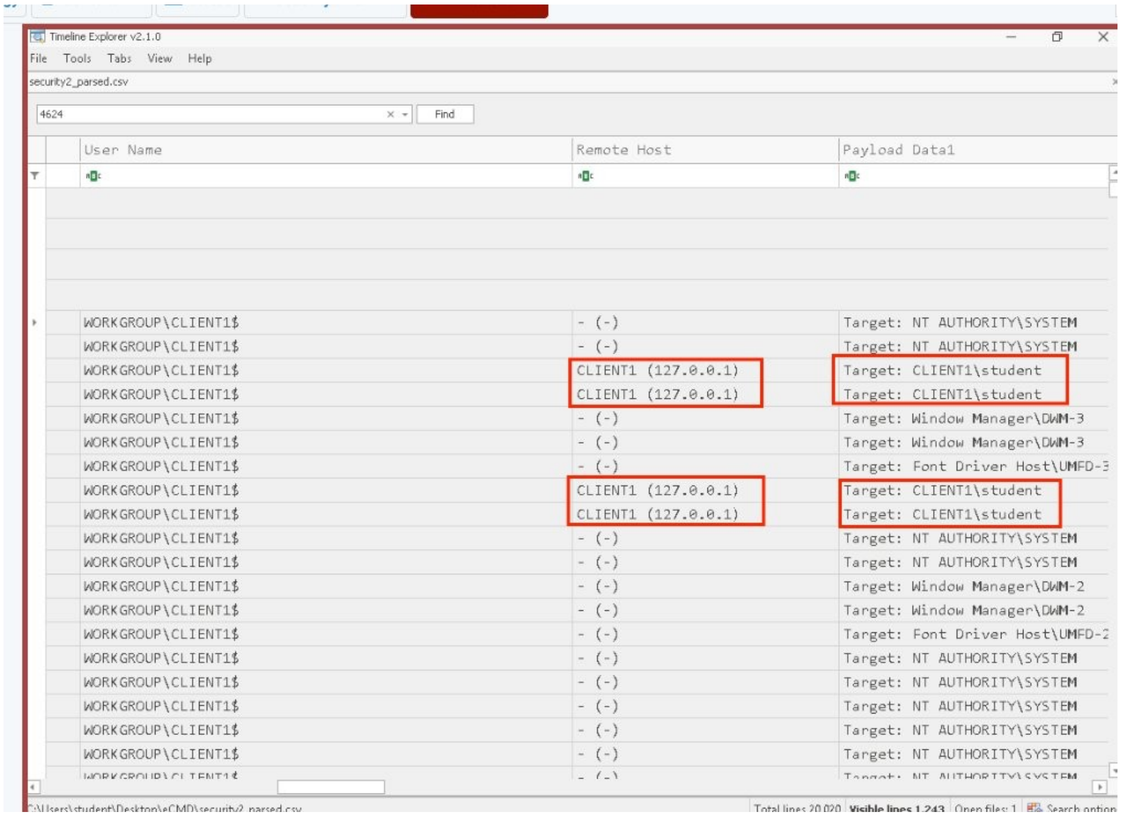

Figure 5. Timeline Explorer filtered to Event ID 4624. The UserName column shows WORKGROUP\CLIENT1$, the RemoteHost column identifies CLIENT1 (127.0.0.1) for interactive sessions, and PayloadData1 confirms the target account as CLIENT1\student.

LogonType values carry important investigative meaning. Type 2 indicates an interactive (console) logon, Type 3 indicates a network logon such as a mapped share, Type 5 indicates a service logon, and Type 10 indicates a remote interactive (RDP) logon. Correlating LogonType with RemoteHost allows an analyst to distinguish between a user sitting at the keyboard, an automated service, and a remote session.

Event ID 4625: Failed Logon

Event ID 4625 is generated when a logon attempt fails. The security2.evtx file contains 2 such events. In Timeline Explorer, filtering for 4625 shows both records with timestamps from June 2024. The Map Description column reads 'Failed logon', confirming the event type at a glance. The UserName column identifies both WORKGROUP\CLIENT1$ and CLIENT1\student as the accounts involved, and the RemoteHost column shows the originating machine. A pattern of repeated 4625 events against the same account in a short timeframe is a classic indicator of a brute-force or password-spray attack.

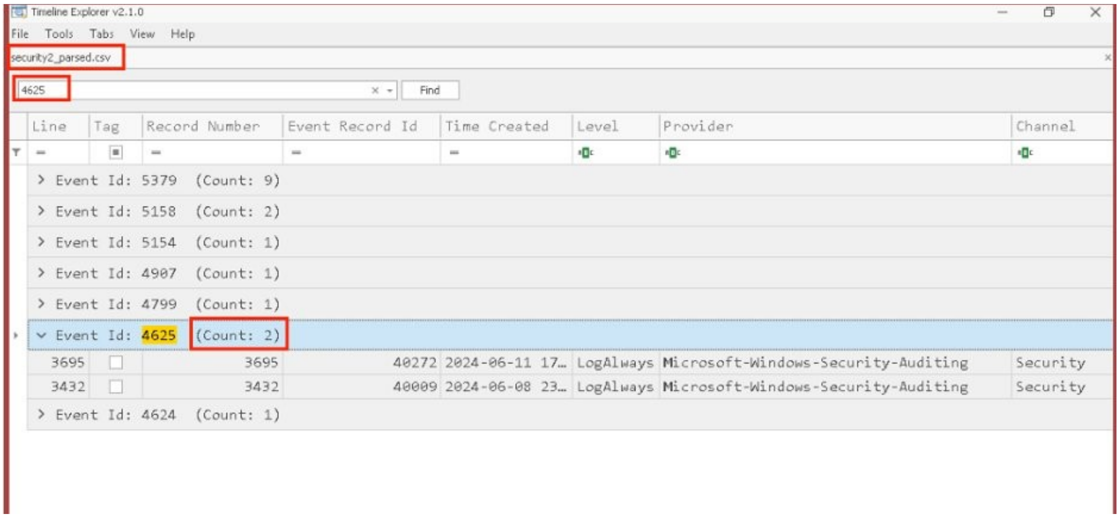

Figure 6. Timeline Explorer filtered to Event ID 4625 (Count: 2). Two failed logon records are visible with timestamps from June 2024.

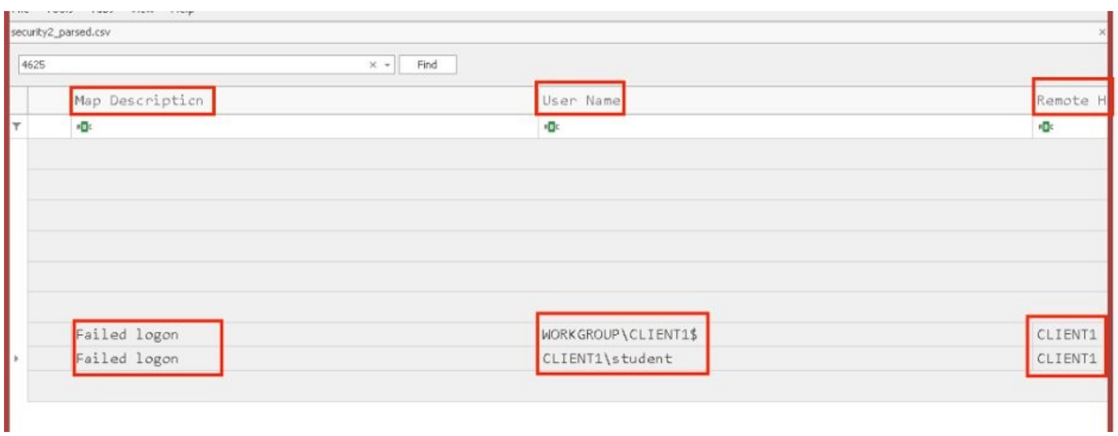

Figure 7. Event ID 4625 detail view showing Map Description 'Failed logon', UserName, and RemoteHost columns. The two affected accounts are WORKGROUP\CLIENT1$ and CLIENT1\student.

Event ID 4634: Logoff

Event ID 4634 is generated when a logon session is terminated. The file contains 24 logoff events. The Map Description column displays 'An account was logged off' for every record. The PayloadData1 column identifies the target account associated with the session that ended, including system processes such as Window Manager\DWM and Font Driver Host\UMFD alongside the primary user account CLIENT1\student. Pairing 4634 events with their corresponding 4624 events using the LogonId field allows an analyst to reconstruct the exact duration of each user session on the machine.

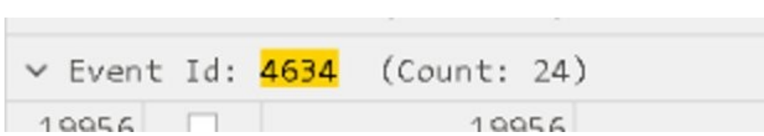

Figure 8. Timeline Explorer showing Event ID 4634 grouped (Count: 24).

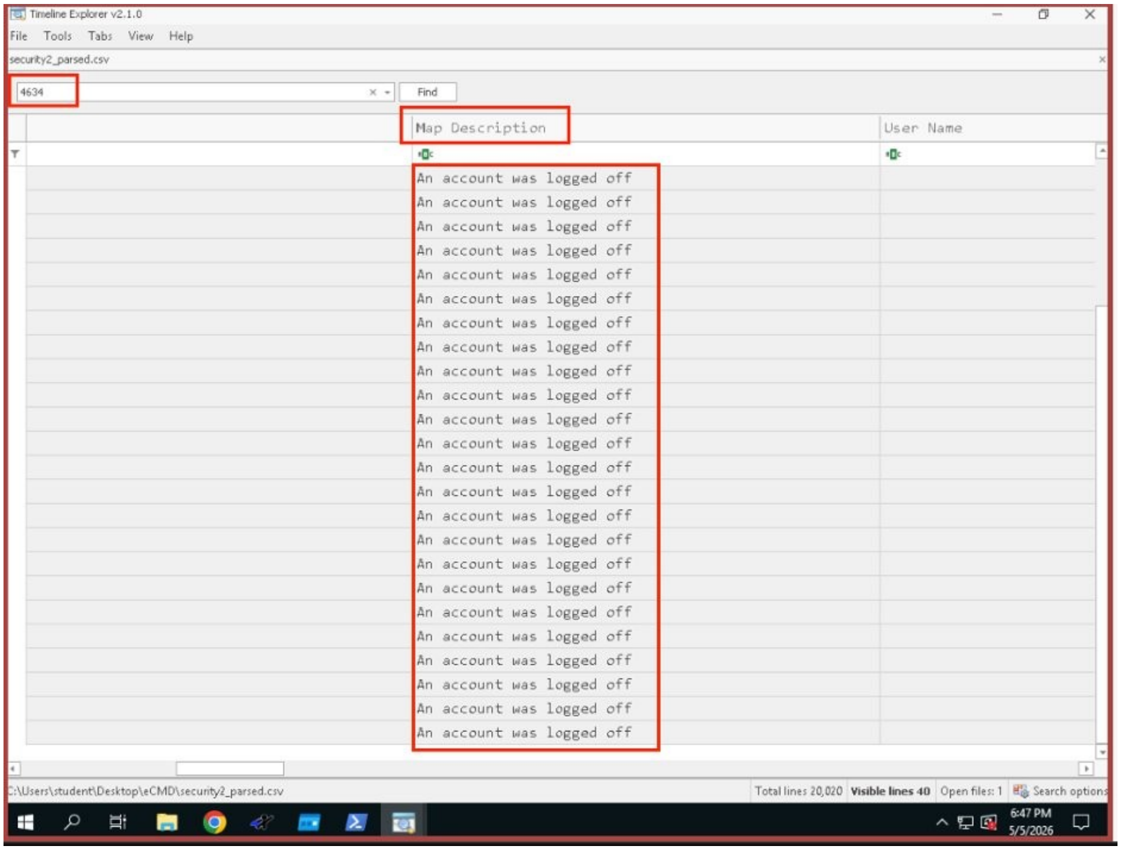

Figure 9. Event ID 4634 Map Description column displaying 'An account was logged off' for all 24 records.

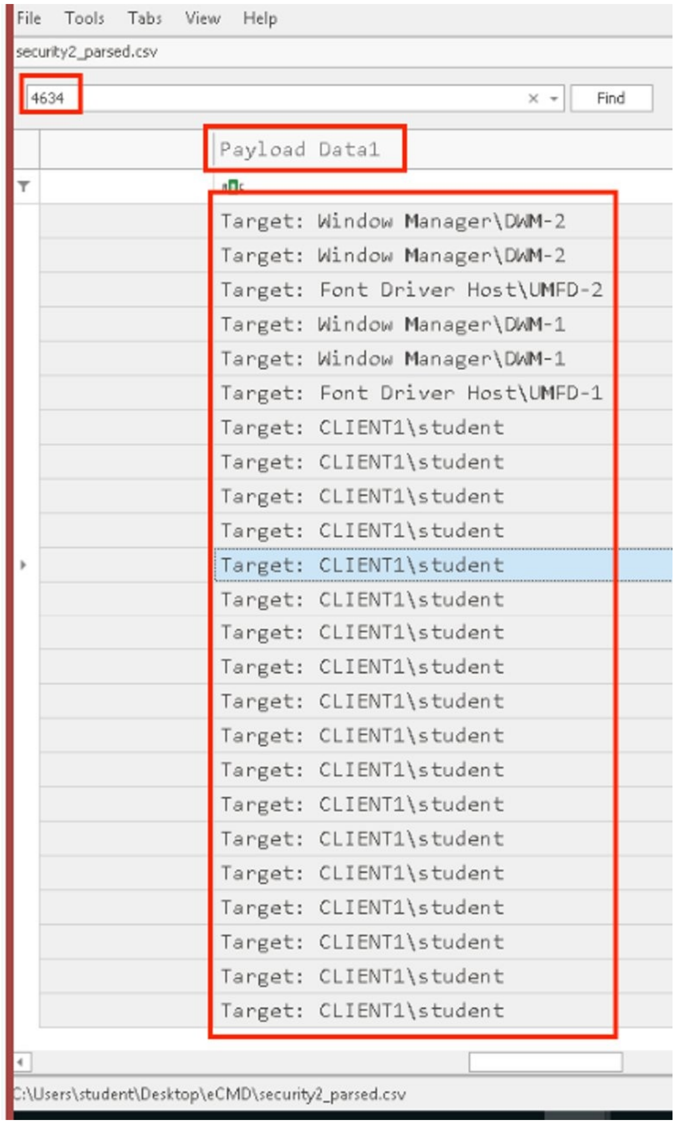

Figure 10. Event ID 4634 PayloadData1 column showing the target accounts whose sessions ended, including CLIENT1\student, Window Manager\DWM, and Font Driver Host\UMFD processes.

Section 2: New Account Creation (Event ID 4720)

Audit Policy Configuration

Event ID 4720 is generated by the Audit account management policy. In secpol.msc, navigate to: Security Settings > Local Policies > Audit Policy > Audit account management.

Double-clicking the policy opens its Properties dialog. Both the Success and Failure checkboxes should be enabled. Enabling Success ensures that Event ID 4720 is logged whenever a new local user account is created, along with related account management events such as 4722 (account enabled), 4724 (password reset attempt), and 4738 (account modified). This single policy covers both Section 2 and Section 3 of this report.

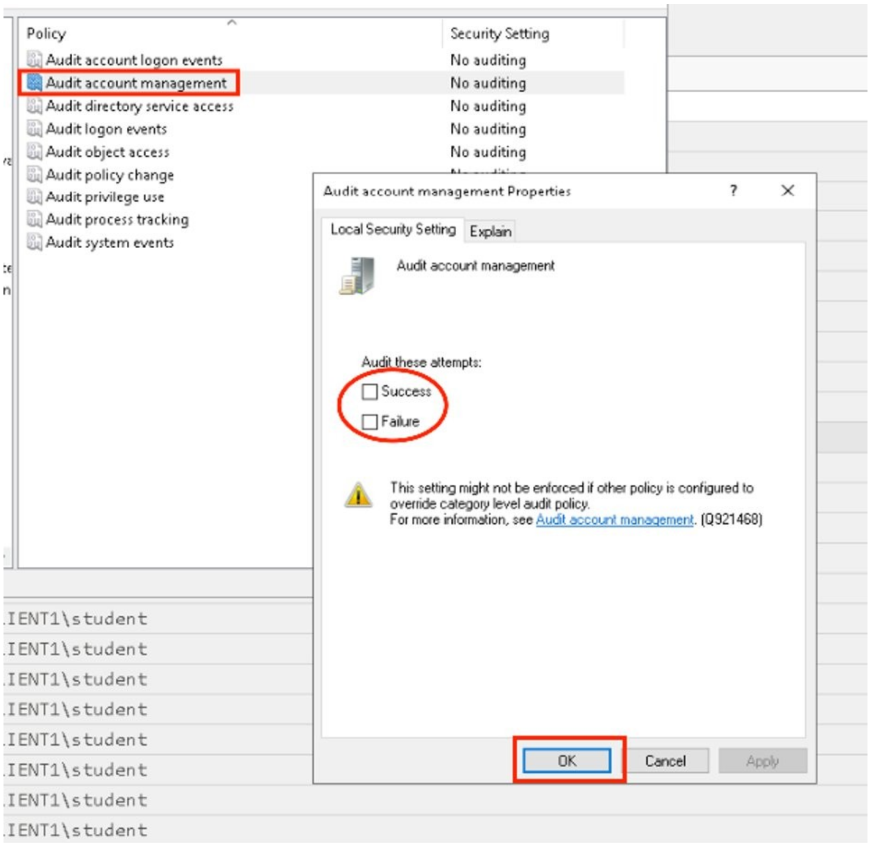

Figure 11. secpol.msc showing Audit account management Properties dialog. Both Success and Failure must be enabled to capture Event IDs 4720 and 4732.

Event ID 4720 in Timeline Explorer

The security2.evtx file contains exactly 1 instance of Event ID 4720. Filtering Timeline Explorer for this event ID reveals a single record. The Map Description column reads 'A new account was created', confirming the event type. The UserName column identifies CLIENT1\student as the account that performed the creation action, and the SID visible in the field is the security identifier assigned to the new account. For an analyst, this event is high-value because legitimate account creation is relatively rare on workstations and any unexpected instance warrants immediate investigation. The timestamp of the event can be correlated with 4624 logon records to determine whether the creator was interactively logged in at the time.

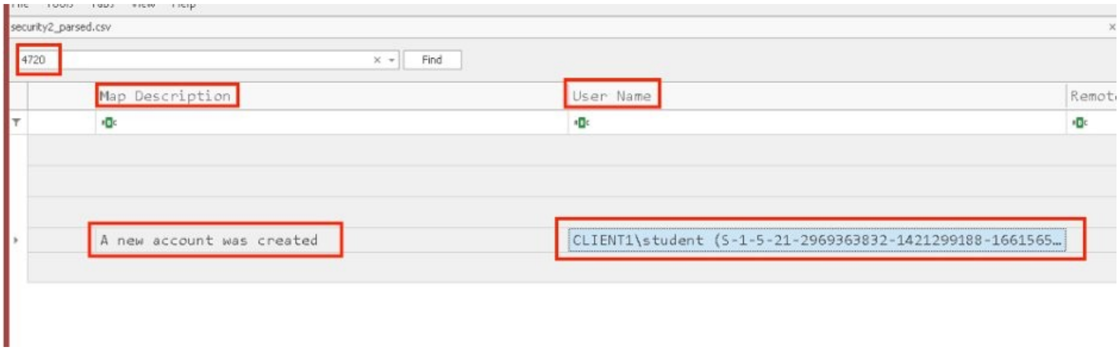

Figure 12. Timeline Explorer filtered to Event ID 4720 (1 record). Map Description confirms 'A new account was created' and UserName identifies CLIENT1\student as the actor.

Section 3: Adding a User to the Local Administrators Group (Event ID 4732)

Audit Policy Configuration

Event ID 4732 is also generated under the Audit account management policy, the same policy configured for Event ID 4720 in Section 2. No additional policy changes are required. As long as Audit account management has Success auditing enabled (Figure 11), Windows will log Event ID 4732 whenever any member is added to a security-enabled local group, including the built-in Administrators group.

This event is among the most critical to monitor in any environment. Adding a standard user account to the local Administrators group grants that account full control over the workstation, including the ability to install software, modify system settings, and access all files. It is a common step in privilege escalation attacks following an initial compromise.

Event ID 4732 in Timeline Explorer

The security2.evtx file contains 2 instances of Event ID 4732. Filtering Timeline Explorer for this event ID reveals both records. The Map Description column reads 'A member was added to a security-enabled local group' for both rows. The UserName column shows CLIENT1\student as the account performing the action in both cases, and the SID column to the right of UserName identifies the specific security principal that was added. An analyst should immediately follow up on this event by examining the surrounding 4624 logon records to confirm who was logged in at the time, and by checking Event ID 4720 to see whether a new account was created immediately before being added to Administrators, which is a common attack pattern.

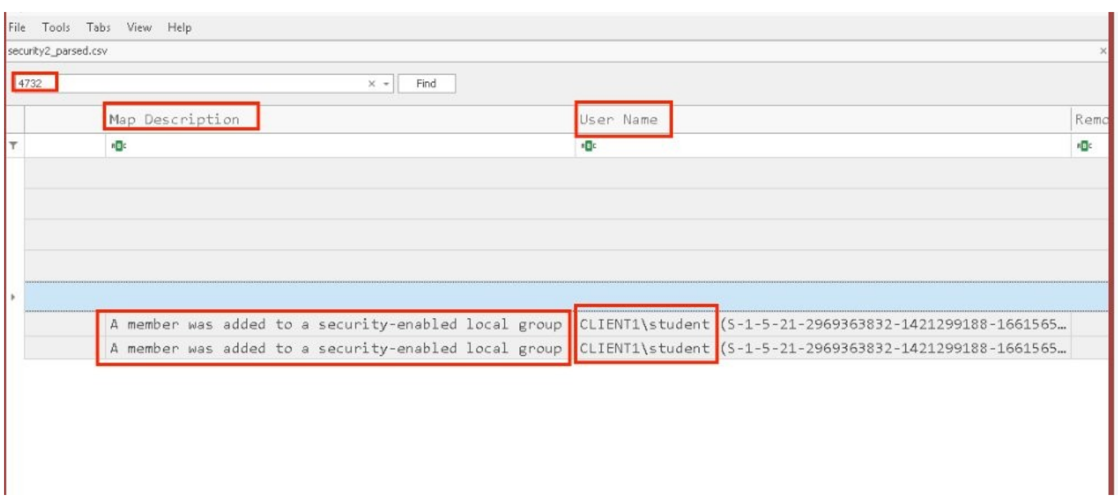

Figure 13. Timeline Explorer filtered to Event ID 4732 (2 records). Map Description reads 'A member was added to a security-enabled local group' and UserName shows CLIENT1\student as the actor for both events.

Summary

This report has demonstrated a complete Windows event log analysis workflow, from policy configuration through to evidence identification in Timeline Explorer. The process begins in secpol.msc, where two audit policies must be enabled: Audit logon events (under Local Policies > Audit Policy) captures authentication activity and generates Event IDs 4624, 4625, and 4634; and Audit account management captures user and group changes, generating Event IDs 4720 and 4732. Without these policies set to Success and Failure, the Security event log will not contain the evidence needed to reconstruct what happened on a target machine.

Once a .evtx file has been collected from a target system, EvtxECmd converts the binary log into a normalized CSV using its Maps system, which translates each event type's unique payload into standardized columns. This single CSV contains all event types in one place, eliminating the need to manage separate output files per event category. The resulting file from security2.evtx contained 20,020 records parsed with 0 errors, with 453 event maps loaded, and was processed in under 15 seconds.

Timeline Explorer provides the analytical layer on top of that CSV. Its column filter boxes allow an analyst to isolate any event ID in seconds, the Map Description column provides a plain-English summary of each event, and columns such as UserName, RemoteHost, and PayloadData1 surface the who, where, and how of each activity. The combination of EvtxECmd's normalization and Timeline Explorer's filtering makes it possible to move from a raw .evtx file to targeted investigative findings in a matter of minutes, a critical capability when responding to a security incident or conducting a forensic examination of a suspect workstation.